

1. Data Sources

Academic Records
(CGPA, Marks,
Backlogs)

Coding Platforms
(DSA, GitHub,
LeetCode)

Resume / Projects
(Skills, Certifications)

Interviews
(Audio, Video,
Feedback)

Co-curricular Activities
(Workshops, Events)

2. Data Ingestion and Mapping

Schema validation, multi-source integration and feature mapping

3. Preprocessing and Feature Engineering

Data Cleaning
(Noise/Outliers)

Missing Value
Handling

Encoding
(Categorical $\rightarrow$ Numerical)

Normalization
(Min-Max / Z-Score)

4. Student Profile Vector (SPV)

$$x_{spv} = [F_1, F_2, \dots, F_n]$$

(n features)

5. Feature Groups in SPV

Academic
(4 features)

Technical
(6 features)

Resume / Career
(4 features)

Behavioral
(4 features)

Co-curricular
(4 features)

6. Prediction and Recommendation Engine

Placement Prediction
(Probability / Category)

Skill Gap Analysis
(Weak Areas Identification)

Personalized Recommendations
(Courses, Skills, Preparation Plan)